NEON SRER Soil & Weather Dataset — Description and Derived Features

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Overview

The dataset is a multivariate, 30-minute soil and weather time series from one NEON site, cleaned and pivoted into one wide table for ML.

Item	Value
Source	US National Ecological Observatory Network (NEON), released data via the NEON API
Site	SRER — Santa Rita Experimental Range, Arizona (semi-arid desert grassland/shrubland)
Time span	Full available record per product ("auto" mode); exact start/end per channel in column_dictionary.txt
Resolution	30-minute averages (totals for precipitation), UTC timestamps
Spatial layout	5 soil plots, sensors at several depths per plot; weather sensors on the site tower
Products	12 continuous sensor products + periodic soil lab chemistry
Labels	None — no soil health or disease labels

Pipeline: neon_soil_downloader.py (download, QC masking, pivot, merge) → drop_empty_columns.py (remove empty channels) → StdErMean columns removed → make_column_dictionary.py (column descriptions, depths, coverage).

Files and structure

Train from wide_30min.parquet; everything else supports or explains it.

File	Contents
wide_30min.parquet / .csv	All products merged on one regular 30-min UTC grid; one column per sensor channel
per_product/*.csv	Same wide format, one file per product, own date range
metadata/_*sensor_positions*.csv	Plot and depth/height (zOffset, m) for each HOR.VER position
metadata/_*variables*.csv	Variable descriptions and units
metadata/_*issueLog*.csv	NEON's known sensor issues and outages
chemistry/*.csv	Periodic lab soil samples: pH, moisture, C and N
column_dictionary.txt / .csv	Readable description, units, plot, depth and coverage per column
download_report.csv / dropped_columns_log.csv	What downloaded, failed, or was removed

Column names follow `<product>_<NEONtable>__<variable>__h<plot>_v<depth index>`. Example: `soil_temp_ST_30_minute__soilTempMean__h001_v502` = soil temperature, plot 1, depth index 502. Weather columns have no plot/depth suffix.

Parameters measured

Twelve sensor products give about 20 physical variables; soil variables repeat across 5 plots and several depths, giving most of the columns.

Product (NEON ID)	Variables (column)	Units	Position
Soil temperature (DP1.00041.001)	soilTempMean	°C	5 plots × up to 9 depths, to ~2 m
Soil water & salinity (DP1.00094.001)	VSWCMean (water content), VSICMean (ion content, salinity proxy)	m^3/m^3; index	5 plots × up to 8 depths
Soil CO_2 (DP1.00095.001)	soilCO2concentrationMean	ppm	5 plots × 3 shallow depths

Product (NEON ID)	Variables (column)	Units	Position
Soil heat flux (DP1.00040.001)	SHFMean	W/m²	5 plots, ~8 cm below surface
IR surface temperature (DP1.00005.001)	bioTempMean	°C	Soil/canopy surface, tower and plots
Air temperature (DP1.00003.001)	tempTripleMean	°C	Tower top
Relative humidity (DP1.00098.001)	RHMean, tempRHMean, dewTempMean	%, °C	Tower
Precipitation (DP1.00006.001)	priPrecipBulk / secPrecipBulk / TFPrecipBulk	mm per 30 min	Gauges (availability varies)
Radiation (DP1.00023.001)	inSWMean, outSWMean, inLWMean, outLWMean	W/m²	Tower
PAR (DP1.00024.001)	PARMean	µmol/m²/s	Tower levels
Wind (DP1.00001.001)	windSpeedMean, windDirMean	m/s, degrees	Tower levels
Barometric pressure (DP1.00004.001)	staPresMean	kPa	Tower
Soil chemistry, periodic (DP1.10086.001)	pH (water, $CaCl_2$), moisture, C, N	various	Lab samples, a few times per year

Exact depths, heights and units per column are in column_dictionary.txt; variable names above may differ slightly by NEON release.

Data quality and caveats

The data is research grade but gappy; trim to a common window and handle NaNs before training.

- **QC masking:** values with NEON final quality flag = 1 were set to NaN. Remaining values passed NEON's automated checks.
- **Gaps:** maintenance, calibration, sensor swaps and late installation leave NaN runs. The coverage % per column is in column_dictionary.txt.

- **Uneven start dates:** products began at different times, so early rows are mostly empty. Pick the window where your chosen channels overlap.
- **Sensor moves:** some soil sensors were reinstalled at slightly different depths; the dictionary flags the latest position and any recorded range.
- **Time zone:** all timestamps are UTC. Arizona local time is UTC−7; Egypt is UTC+2 (UTC+3 in summer).
- **No labels:** there is no soil health or disease ground truth. Use the data for forecasting, anomaly detection, pretraining or pipeline testing.
- **Salinity index:** VSICMean is an ion-content index from the water sensor, not a calibrated EC in dS/m.
- **Chemistry:** pH and C/N are sparse lab points, not a time series.

Derived and calculated features

The raw channels support about 40 physically meaningful derived features, from vapour pressure deficit to soil water storage and thermal diffusivity. T = temperature in °C, P = pressure in kPa, θ = volumetric water content, z = depth in m.

Atmosphere and humidity

Feature	Calculation	Inputs	Why it matters
Saturation vapour pressure e_s	FAO-56 formula below	tempTripleMean	Basis for all humidity features
Actual vapour pressure e_a	$e_s \times RH / 100$	RHMean, air temp	Absolute moisture in air
Vapour pressure deficit (VPD)	$e_s - e_a$	RH, air temp	Drives evaporation and plant water stress
Specific humidity q	$0.622\, e_a / (P - 0.378\, e_a)$	RH, temp, staPresMean	Pressure-independent moisture measure
Dew point depression	T − T_dew	tempRHMean, dewTempMean	Condensation and dew risk on leaves
Psychrometric constant γ	$0.000665 \times P$	staPresMean	Needed for evapotranspiration
Growing degree days	Σ max(0, T_daily_mean − T_base), base often 10 °C	air temp	Crop and pathogen development stage

Formula: $e_s(T) = 0.6108 \times \exp(17.27\, T / (T + 237.3))$, in kPa

Radiation and surface energy balance

Feature	Calculation	Inputs	Why it matters
Net shortwave	inSW − outSW	inSWMean, outSWMean	Energy absorbed by the surface
Albedo	outSW / inSW (daytime only, inSW > 50 W/m²)	inSW, outSW	Changes with soil wetness and vegetation cover
Net longwave	inLW − outLW	inLWMean, outLWMean	Night cooling, cloud effect
Net radiation R_n	net SW + net LW	all 4 radiation channels	Main energy input to soil and plants

Feature	Calculation	Inputs	Why it matters
Radiometric surface temperature	(outLW / (εσ))^0.25 − 273.15, ε ≈ 0.95	outLWMean	Cross-check against IR surface temperature
Available energy	R_n − G	R_n, SHFMean	Energy left for evaporation and heating air
Ground heat fraction	G / R_n	R_n, SHFMean	Indicates soil dryness and vegetation cover
PAR fraction	PAR / inSW	PARMean, inSW	Light quality for photosynthesis
Day/night flag, solar elevation	from timestamp + site lat/lon, or inSW > 10 W/m²	timestamp, inSW	Separates day and night behaviour
Clearness index	inSW / extraterrestrial radiation	inSW, solar geometry	Cloud cover proxy

Evapotranspiration

Feature	Calculation	Inputs	Why it matters
Wind at 2 m, u_2	u_z × 4.87 / ln(67.8 z − 5.42)	windSpeedMean, sensor height z	Standardises wind for ET formulas
Reference ET_0 (FAO-56 Penman–Monteith, hourly)	formula below, summed to daily	R_n, G, T, VPD, u_2, γ	Atmospheric water demand; irrigation scheduling
Priestley–Taylor ET	1.26 × Δ / (Δ + γ) × (R_n − G) / λ	R_n, G, T, P	Simpler ET when wind is missing

Formula: $ET_0 = [\ 0.408\ \Delta\ (R_n - G) + \gamma \times 37 / (T + 273) \times u_2\ (e_s - e_a)\ ] / [\ \Delta + \gamma\ (1 + 0.34\ u_2)\ ]$

Hourly form, with R_n and G in MJ/m²/h (W/m² × 0.0036); Δ is the slope of the e_s curve; the 0.34 coefficient becomes 0.24 by day and 0.96 by night in the ASCE hourly standard.

Soil water and water balance

Feature	Calculation	Inputs	Why it matters
Profile water storage S	$\Sigma \theta_i \times \Delta z_i \times 1000$ (mm)	VSWCMean at all depths, layer thicknesses	Total water available in the root zone
Storage change ΔS	$S(t) - S(t-1)$	S	Wetting and drying events
Water balance residual	$P - \Delta S \approx ET +$ drainage + runoff	precip, S	Estimates water loss rate
Relative soil wetness	$(\theta - \theta_{min}) / (\theta_{max} - \theta_{min})$, per depth	VSWCMean history	Scale-free; transfers across sensors and sites
Plant-available water	$(\theta - \theta_{wp}) / (\theta_{fc} - \theta_{wp})$	VSWCMean + field capacity and wilting point (megapit texture)	Crop water stress
Wetting-front arrival lag	time from rain start to θ rise at each depth	precip, VSWCMean by depth	Infiltration speed; soil structure
Drydown time constant τ	fit $\theta(t) = \theta_{res} + A e^{(-t/\tau)}$ after rain	VSWCMean	Drainage and evaporation behaviour
Antecedent precipitation index	$API_t = k \times API_{(t-1)} + P_t$, $k \approx 0.9$ per day	precip	Memory of recent rain
Days since last rain, cumulative rain	event counters, rolling sums (24 h, 7 d, 30 d)	precip	Simple, strong predictors
Pore-water salinity proxy	VSIC / θ	VSICMean, VSWCMean	Salt concentrates as soil dries

Soil temperature and heat

Feature	Calculation	Inputs	Why it matters
Vertical gradient dT/dz	$(T_1 - T_2) / (z_2 - z_1)$	soilTempMean at adjacent depths	Direction of heat flow
Diurnal range per depth	daily max − daily min	soilTempMean	Damps with depth and with wetter soil
Phase lag between depths	time of daily max at depth 2 − depth 1	soilTempMean	Heat transport speed
Thermal diffusivity κ	amplitude method below (or phase-lag method)	daily amplitudes at two depths	Changes with soil moisture; a soil state indicator
Surface–soil difference	IR surface temp − shallowest soil temp	bioTempMean, soilTempMean	Surface heating and drying
Surface–air difference	IR surface temp − air temp	bioTempMean, tempTripleMean	Canopy water stress (crop water stress index idea)

Formula: $\kappa = \omega\,(z_2 - z_1)^2 / (\,2\,[\,\ln(A_1 / A_2)\,]^2\,)$, with $\omega = 2\pi / 86400$ s and A = daily temperature amplitude at each depth

Soil CO_2 and biological activity

Feature	Calculation	Inputs	Why it matters
CO_2 gradient dC/dz	between the 3 measured depths	soilCO2concentrationMean	Direction and strength of CO_2 production
Soil CO_2 efflux (flux-gradient)	$F = -D_s\,dC/dz$, D_s from air-filled porosity (Millington–Quirk)	CO_2 profile, θ, porosity (megapit bulk density)	Soil respiration estimate
Temperature sensitivity Q_{10}	fit $C = a \times Q_{10}^{\wedge}(T/10)$	CO_2, soil temp	Biological activity response to heat
Rain pulse response	CO_2 rise after rain on dry soil	CO_2, precip, θ	Microbial activation (Birch effect)

Feature	Calculation	Inputs	Why it matters
Moisture–CO_2 interaction	CO_2 binned by θ and T	CO_2, θ, T	Separates water and temperature limits

Encodings, aggregation and generic time-series features

Feature	Calculation	Why it matters
Wind components u, v	u = −ws sin(dir), v = −ws cos(dir)	Removes the 0°/360° jump in direction
Cyclic time	sin/cos of hour-of-day and day-of-year	Lets models learn daily and seasonal cycles
Cross-plot mean and spread	mean and std over the 5 plots per depth	Fewer columns; robust to one failed sensor
Rolling statistics	mean, std, min, max over 3 h, 24 h, 7 d	Smooths noise; captures trends
Slopes and differences	linear-fit slope over a window; first difference	Rate of drying, warming, salinity build-up
Daily aggregates	daily min, max, mean, range	Reduces 48 steps/day to compact features
Lags	values at t−1, t−48 (1 day), t−336 (1 week)	Persistence and memory
Climatology anomalies	z-score vs same hour and day-of-year	Flags unusual conditions, sensor faults
Cross-correlation lags	lag of max correlation between depths or drivers	Response time of soil to weather

Suggested uses and next steps

Without labels, the dataset is best for forecasting, anomaly detection and pretraining, then transfer to the labelled 26-sensor soil health data.

Use	Target or method	Key features
Soil moisture forecasting	θ or profile storage S, 1–48 h ahead	precip, API, ET_0, VPD, lags
Soil temperature forecasting	soilTempMean per depth	radiation, air temp, θ, cyclic time
Anomaly / sensor-fault detection	autoencoder or isolation forest on residuals	climatology anomalies, cross-plot spread
Self-supervised pretraining	masked-channel or next-step prediction	all channels; fine-tune on own labels
Stress-event detection (weak labels)	rule-based labels, e.g. low relative wetness + high VPD	relative wetness, VPD, surface–air difference
Pipeline and edge testing	run the full feature code on Raspberry Pi / Jetson	rolling stats, slopes, ET_0 (all cheap to compute online)

To transfer to your own sensors, keep only features both datasets can produce, and prefer scale-free versions (relative wetness, anomalies, slopes) over raw values, since probes, soils and climate differ.

- [ ] Pick a training window where the chosen channels have more than 80% coverage
- [ ] Average plots per depth, or keep the plot with the best coverage
- [ ] Implement the derived features in one feature module shared by training and the edge device
- [ ] Pull field capacity, wilting point and bulk density from the megapit data for plant-available water and CO_2 flux
- [ ] Map each of the 26 own sensors to its closest NEON channel or derived feature